

AI Course

Capstone Project Work Breakdown Structure

For students

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Capstone Project Work Breakdown Structure (WBS)

Today	10/4/2026
Project Team Name	Group 6 (Cogni)
Project Topic	Bayan: Arabic text simplification assistant for readers with dyslexia

	Task	Start date	End date	Responsibility	Deliverable	Task Status	W1					W2					W3					W4				
							D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20
							9/16	9/17	9/18	9/19	9/20	9/21	9/22	9/23	9/24	9/25	9/26	9/27	9/28	9/29	9/30	10/1	10/2	10/3	10/4	10/5
1.	Project initiation and planning	16 Sep	22 Sep	Marwan, Abdulrahman		Done																				
	1.1. Project brief and reference architecture	16 Sep	16 Sep	Abdulrahman	Brief, architecture report	Done																				
	1.2. Verified literature and benchmark references	16 Sep	17 Sep	Marwan	Reference library	Done																				
	1.3. Action Plan draft	17 Sep	19 Sep	Marwan	Action Plan draft	Done																				
	1.4. Execution plan: 13 milestones with risks and mitigations	16 Sep	16 Sep	Abdulrahman	Execution plan	Done																				
	1.5. Final Action Plan, every claim labelled measured / decided / planned (#25)	21 Sep	22 Sep	Marwan	Final Action Plan	Done																				
2.	Project management	17 Sep	4 Oct	Marwan		Done																				
	2.1. Team setup: repository, workflow and roles	17 Sep	18 Sep	Marwan	Repository, team guidance	Done																				
	2.2. Risk register and data contingency plan	18 Sep	19 Sep	Marwan	Risk register	Done																				
	2.3. Product form decision record: Android plugin, on device (#9)	20 Sep	20 Sep	Marwan, Mohammed	Decision record	Done																				
	2.4. Task issues with handoffs, and pull-request reviews	19 Sep	4 Oct	Marwan	GitHub issues, reviews	Done																				
	2.5. Mentoring sessions: progress shown (23 and 30 Sep)	23 Sep	30 Sep	Marwan	Progress reports	Done																				
3.	Data	16 Sep	4 Oct	Abdulrahman, Marwan, Ahmed		Done																				
	3.1. BAREC readability analysis	16 Sep	16 Sep	Abdulrahman	Analysis report	Done																				
	3.2. Reading-level classifier, ONNX int8 (CAMeL AraBERT) (#17, #20)	16 Sep	21 Sep	Abdulrahman	Level classifier	Done																				
	3.3. Equivalence validator and easy-band gate	16 Sep	20 Sep	Abdulrahman	Validator, gate	Done																				
	3.4. Synthetic data pipeline A (DeepSeek on BAREC)	16 Sep	16 Sep	Abdulrahman	Pipeline A	Done																				
	3.5. Pipeline A 10% run: 3,010 sentences scored, 5,054 clean pairs	16 Sep	17 Sep	Abdulrahman	Generated pairs v1	Done																				
	3.6. Synthetic data pipeline B (Gemini and Gemma): 5,618 checked pairs	18 Sep	19 Sep	Marwan	Synthetic dataset v1	Done																				
	3.7. Shared data format specification	19 Sep	19 Sep	Marwan, Abdulrahman	Format spec	Done																				
	3.8. Word-difficulty lexicon (24,458 words)	20 Sep	20 Sep	Abdulrahman	Lexicon	Done																				
	3.9. Validation module (scripts/validation.py) (#10, #11)	19 Sep	20 Sep	Abdulrahman, Marwan	Validation module	Done																				
	3.10. Human review set (500 pairs) and annotation tool	20 Sep	20 Sep	Abdulrahman	Review set, tool	Done																				
	3.11. SAMER corpus in the shared format (#5)	19 Sep	21 Sep	Ahmed	SAMER dataset	Done																				
	3.12. Leakage audit: 9.3% of Baseet in the locked test sets; leakage gate	21 Sep	21 Sep	Marwan	Audit, clean data	Done																				
	3.13. Re-labelled synthetic export (#12)	20 Sep	25 Sep	Abdulrahman	Re-labelled export	Done																				
	3.14. Model-2 training data: 22,733 tagged, meaning-filtered pairs	22 Sep	24 Sep	Marwan	Training set	Done																				
	3.15. Synthetic corpus v0 exported and published: 13,072 pairs (#8, #38, #42)	25 Sep	30 Sep	Abdulrahman	Corpus v0 (Hub)	Done																				
3.16. Corpus v1: Gemma 4 generator, Qwen judge, 14,975 rows, data card (#44, #45)	28 Sep	4 Oct	Abdulrahman	Corpus v1, data card	Done																					
4.	Models	19 Sep	4 Oct	Ahmed, Sanad, Abdulrahman, Marwan		Done																				
	4.1. Diacritization benchmark; Libtashkeel selected (#6, #23)	19 Sep	21 Sep	Sanad	Benchmark report	Done																				
	4.2. Read-aloud model selection: Nabra-7M-Distill	21 Sep	22 Sep	Marwan	Selection	Done																				
	4.3. Model 1: AraT5v2 on SAMER, trained and documented (#5, #22)	19 Sep	23 Sep	Ahmed	Model 1	Done																				
	4.4. Model 1 diagnosis on the locked test sets	22 Sep	23 Sep	Marwan	Diagnosis	Done																				
	4.5. Model 2 on AraT5v2 (#28, #40)	24 Sep	25 Sep	Ahmed	Model 2 (AraT5v2)	Done																				
	4.6. Model 2 on AraBART, and the speed benchmark (#29, #35)	24 Sep	25 Sep	Sanad	Model 2 (AraBART)	Done																				
	4.7. Meaning judge: Gemma 4 E2B, four heads, three variants (#47, #49)	30 Sep	4 Oct	Abdulrahman	Meaning judge (Hub)	Done																				
4.8. Model 3: AraT5v2 on corpus v1, int8 bundle (#48, #53)	2 Oct	4 Oct	Abdulrahman	Model 3	Done																					
5.	Evaluation	19 Sep	4 Oct	Abdul Majid, Marwan		Done																				
	5.1. Scoring harness and locked test sets (#7, #15, #16, #18)	19 Sep	21 Sep	Abdul Majid, Marwan	Scripts, manifest	Done																				
	5.2. Copy baselines, locked-set converter, generate-and-score path (#19, #24, #26)	21 Sep	22 Sep	Abdul Majid	Baselines	Done																				
	5.3. Model-2 scoring tools (#33, #34)	24 Sep	24 Sep	Marwan	Scoring tools	Done																				
	5.4. BayanBench v1: behaviour tracks, document split; model 2 scored on test	26 Sep	28 Sep	Marwan	BayanBench v1	Done																				
	5.5. BayanBench v2: open meaning scorer (Gemma 4 31B); all systems scored (#48)	29 Sep	30 Sep	Marwan, Abdulrahman	BayanBench v2	Done																				
	5.6. Human rating round: 290 tasks, five raters and a reader with dyslexia (#48)	30 Sep	3 Oct	Marwan, Sanad, Ahmed, Moh	Ratings, scorer check	Done																				
	5.7. BayanBench v2.0 frozen; test split scored (#55)	3 Oct	4 Oct	Marwan, Abdulrahman	BayanBench v2.0	Done																				
5.8. Model 3 ship gate and decision: an option next to model 2 (#56, #59)	4 Oct	4 Oct	Marwan, Abdulrahman	Decision	Done																					
6.	Application	20 Sep	4 Oct	Mohammed, Marwan		Done																				
	6.1. PROCESS_TEXT plugin and result sheet (#14, #21)	20 Sep	21 Sep	Mohammed	Android plugin	Done																				
	6.2. Model download, result states, settings, typography (#31, #37)	24 Sep	30 Sep	Mohammed	App features	Done																				
	6.3. Compose app with Material 3 Expressive (#37)	26 Sep	30 Sep	Marwan	App	Done																				
	6.4. Read-aloud with word highlighting (Nabra through sherpa-onnx)	27 Sep	30 Sep	Marwan	Read-aloud	Done																				
	6.5. Overlay, floating reader, text-step guards, Bayan design system (#54)	1 Oct	4 Oct	Marwan	App release	Done																				
6.6. Beam search as a per-model decoding option (#61, #62)	4 Oct	4 Oct	Marwan	Decoder	Done																					
7.	On-device model	24 Sep	4 Oct	Marwan		Done																				
	7.1. ONNX expprt, int8 quantization, on-device engine (#32)	24 Sep	26 Sep	Marwan	Model bundles	Done																				
	7.2. Phone latency: AraT5v2 (26 Sep), AraBART (3 Oct), beam search (4 Oct)	26 Sep	4 Oct	Marwan	Measurements	Done																				
8.	Final deliverables	29 Sep	5 Oct	Marwan, Abdulrahman		In-progress																				
	8.1. Final report (#52)	29 Sep	5 Oct	Marwan, Abdulrahman	Final report	In-progress																				
	8.2. Presentation deck (#50)	2 Oct	5 Oct	Abdulrahman	Slide deck	In-progress																				
	8.3. Coding reference: repository and README (#60)	4 Oct	5 Oct	Marwan	Repository	In-progress																				
	8.4. Demo video			Marwan	Demo video																					
8.5. Final pitch (Tue 6 Oct)			Marwan, Abdulrahman	Pitch																						